

Changing Case in a String with Methods

One of the simplest tasks you can do with strings is change the case of the words in a string. Look at the following code, and try to determine what's happening:

```
name.py name = "ada lovelace"  
print(name.title())
```

Save this file as *name.py* and then run it. You should see this output:

```
Ada Lovelace
```

In this example, the variable *name* refers to the lowercase string "ada lovelace". The method `title()` appears after the variable in the `print()` call. A *method* is an action that Python can perform on a piece of data. The dot (`.`) after *name* in `name.title()` tells Python to make the `title()` method act on the variable *name*. Every method is followed by a set of parentheses, because methods often need additional information to do their work. That information is provided inside the parentheses. The `title()` function doesn't need any additional information, so its parentheses are empty.

The `title()` method changes each word to title case, where each word begins with a capital letter. This is useful because you'll often want to think of a name as a piece of information. For example, you might want your program to recognize the input values *Ada*, *ADA*, and *ada* as the same name, and display all of them as *Ada*.

Several other useful methods are available for dealing with case as well. For example, you can change a string to all uppercase or all lowercase letters like this:

```
name = "Ada Lovelace"  
print(name.upper())  
print(name.lower())
```

This will display the following:

```
ADA LOVELACE  
ada lovelace
```

The `lower()` method is particularly useful for storing data. You typically won't want to trust the capitalization that your users provide, so you'll convert strings to lowercase before storing them. Then when you want to display the information, you'll use the case that makes the most sense for each string.

Using Variables in Strings

In some situations, you'll want to use a variable's value inside a string. For example, you might want to use two variables to represent a first name and